1.01 Complex Numbers: Algebra

Worked Solutions 1–48

1 a)

$$\sqrt{-9} = \sqrt{9} \times \sqrt{-1} = 3i$$

1 b)

$$\sqrt{-2} = \sqrt{2} \times \sqrt{-1} = \sqrt{2}i$$

1 c)

$$2\sqrt{-4} = 2 \times \sqrt{4} \times \sqrt{-1} = 4i$$

1 d)

$$-3\sqrt{-2} = -3 \times \sqrt{2} \times \sqrt{-1} = -3\sqrt{2}i$$

2)

$$\sqrt{-a} \times \sqrt{-b} = \sqrt{a}i \times \sqrt{b}i = \sqrt{a \times b}i^2 = -\sqrt{ab}$$

3 a)

$$i^2 = -1$$

3 b)

$$i^3 = i \times i^2 = i \times -1 = -i$$

3 c)

$$i^4 = i^2 \times i^2 = -1 \times -1 = 1$$

3 d)

$$i^5 = i \times (i^2)^2 = i \times (-1)^2 = i$$

3 e)

$$i^{40} = (i^2)^{20} = (-1)^{20} = 1$$

3 f)

$$i^{41} = i \times (i^2)^{20} = i \times (-1)^{20} = i$$

3 g)

$$i^{99} = i \times (i^2)^{49} = i \times (-1)^{49} = -i$$

4 a)

$$i^{4n} = (i^2)^{2n} = (-1)^{2n} = 1$$

4 b)

$$i^{4n+1} = i \times i^{4n} = i \times (i^2)^{2n} = i \times (-1)^{2n} = i$$

4 c)

$$i^{4n+2} = i^2 \times i^{4n} = i^2 \times (i^2)^{2n} = -1 \times (-1)^{2n} = -1$$

4 d)

$$i^{4n+3} = i^3 \times i^{4n} = i \times i^2 \times (i^2)^{2n} = i \times -1 \times (-1)^{2n} = -i$$

4 e)

$$i^{4n+17} = i \times i^{4n+16} = i \times (i^2)^{2n+8} = i \times (-1)^{2n+8} = i$$

4 f)

$$i^{4n+26} = i^2 \times i^{4n+24} = -1 \times (i^4)^{n+6} = -1 \times (1)^{n+6} = -1$$

5)

$$i + i^2 + i^3 + i^4 = i - 1 - i + 1 = 0$$
$$i + i^2 + i^3 + \cdots + i^{1000} = 250(i - 1 - i + 1) = 0$$

6 a)

$$(1 - i)^2 = (1 - i)(1 - i) = 1 - 2i + (-i)^2 = 1 - 2i - 1 = -2i$$

6 b)

$$(1-i)^3 = (1-i)(1-i)(1-i) = -2i(1-i) = -2i + 2i^2 = -2 - 2i$$

6 c)

$$(1-i)^2 = (1-i)(1-i) = 1 - 2i + (-i)^2 = 1 - 2i - 1 = -2i$$
$$(1-i)^4 = (1-i)^2(1-i)^2 = -2i \times -2i = 4 \times -1 = -4$$

6 d)

$$(1+i)^2 = (1+i)(1+i) = 1 + 2i + i^2 = 2i$$

6 e)

$$(1+i)^3 = (1+i)(1+i)(1+i) = 2i(1+i) = 2i + 2i^2 = -2 + 2i$$

6 f)

$$(1+i)^2 = (1+i)(1+i) = 1 + 2i + i^2 = 2i$$
$$(1+i)^4 = (1+i)^2(1+i)^2 = 2i \times 2i = 4 \times -1 = -4$$

7 a)

$$\sqrt{-12} \times \sqrt{-27} = \sqrt{3 \times 4 \times -1} \times \sqrt{3 \times 9 \times -1}$$
$$= 2\sqrt{3}i \times 3\sqrt{3}i = 6 \times 3 \times -1 = -18$$

7 b)

$$(2i)^3 = 8i^3 = -8i$$

7 c)

$$i^2(i+3) = -1(i+3) = -i - 3$$

7 d)

$$i + 4i^5 = i + 4(-1)(-1)i = i + 4i = 5i$$

7 e)

$$(i - 3i^3)(1 - 2i^4) = (i + 3i)(1 - 2) = (4i)(-1) = -4i$$

7 f)

$$\begin{aligned}(2i^6 - i^9)^2 &= (2(-1)^3 - i(-1)^4)^2 \\ &= (-2 - i) = 4 + 4i + i^2 = 3 + 4i\end{aligned}$$

8)

$$\left(i^{12} + \left(\frac{1}{i}\right)^{18}\right)^3 = \left((-1)^6 + \left(\frac{1}{-1}\right)^9\right)^3 = (1-1)^3 = 0$$

9)

$$\left(i^{30} - \left(\frac{1}{i}\right)^{12}\right)^4 = \left((-1)^{15} - \left(\frac{1}{-1}\right)^6\right)^4 = (-1-1)^4 = 16$$

10 a)

$$2 - 3i + (1 + i) = 3 - 2i$$

10 b)

$$2 - 3i + (5 - 2i) = 7 - 5i$$

10 c)

$$5 - 2i - (2 - 3i) = 3 + i$$

10 d)

$$2(1 + i) - (2 - 3i) = 2 + 2i - 2 + 3i = 5i$$

10 e)

$$5(2 - 3i) + 3(1 + i) = 10 - 15i + 3 + 3i = 13 - 12i$$

11)

$$(2 + 3i)^2 = (2 + 3i)(2 + 3i) = 4 + 12i - 9 = -5 + 12i \quad \text{as req.}$$

12)

$$\begin{aligned}(-1 + 5i)^3 &= (-1 + 5i)(-1 + 5i)(-1 + 5i) \\ &= (1 - 10i - 25)(-1 + 5i) \\ &= (-24 - 10i)(-1 + 5i) = 24 + 10i - 120i + 50 \\ &= 74 - 110i \quad \text{as req.}\end{aligned}$$

13 a)

$$\mathrm{Re}(z) = 2 + 3x \quad \text{and} \quad \mathrm{Im}(z) = -y$$

13 b)

$$\mathrm{Re}(z) = x - 2x = -x \quad \text{and} \quad \mathrm{Im}(z) = -2 - 3y$$

13 c)

$$\mathrm{Re}(z) = 5x + 2x = 7x \quad \text{and} \quad \mathrm{Im}(z) = -5y + 2y = -3y$$

14 a)

$5x + 2y = 25$ and $3x + 4y = 29$

Solving simultaneously: $7x = 21 \ \therefore x = 3$ and $y = 5$

14 b)

$2x + 8y = -20$ and $x - 4y = 14$

Solving simultaneously: $4x = 8 \ \therefore x = 2$ and $y = -3$

14 c)

$$(-3x + 2ix) + (2y + 7iy) = -3x + 2y + (2x + 7y)i$$
$$-3x + 2y + (2x + 7y)i = -14 - 74i$$

$-3x + 2y = -14$ and $2x + 7y = -74$

Solving simultaneously: $25y = -250 \ \therefore x = -2$ and $y = -10$

14 d)

$x + 3x^2 = 2$ and $9x - y = 1$

$$3x^2 + x - 2 = 0 \Rightarrow x = \frac{-1 \pm \sqrt{1 + 24}}{6} = \frac{2}{3} \text{ or } -1$$

$\therefore x = \frac{2}{3}$ and $y = 5$ or $x = -1$ and $y = -10$

15)

$$z = (3 + 7i)(a + 2i) = 3a + 6i + 7ai - 14$$
$$\mathrm{Re}(z) = 3a - 14 = 0 \ \therefore a = \frac{14}{3}$$

16)

$$z = (2 + i)(a + bi) = 2a + ai + 2bi - b$$

$\mathrm{Re}(z) = 0 \ \therefore 2a - b = 0$ so $b = 2a$

$$z = 2a + ai + 2(2a)i - 2a = 5ai$$

17)

$$(a-bi)(a+3bi) = a^2+3abi-abi+3b^2 = a^2+3b^2+2abi$$

$a^2+3b^2 = 52$ and $2ab = 16$

$$\therefore a = \frac{8}{b} \text{ so } \left(\frac{8}{b}\right)^2 + 3b^2 = 52$$

$$\frac{64}{b^2} + 3b^2 = 52 \Rightarrow 64 + 3b^4 = 52b^2 \Rightarrow 3b^4 - 52b^2 + 64 = 0$$

$$\therefore b^2 = \frac{4}{3} \text{ or } b^2 = 16$$

Since b is an integer, $b = \pm 4$

$\therefore a = -2$ and $b = -4$ or $a = 2$ and $b = 4$

18 a)

$$\begin{aligned}(z+1-2i)^2 &= (z+1-2i)(z+1-2i)\\ &= z^2+z-2iz+z+1-2i-2iz-2i-4\\ &= z^2+2z-4iz-3-4i\\ &= z^2+(2-4i)z+(-3-4i)\end{aligned}$$

18 b)

$$\begin{aligned}(z-2+3i)^2 &= (z-2+3i)(z-2+3i)\\ &= z^2-2z+3iz-2z+4-6i+3iz-6i-9\\ &= z^2-4z+6iz-5-12i\\ &= z^2+(-4+6i)z+(-5-12i)\end{aligned}$$

18 c)

$$\begin{aligned}&(z+1+2i)(z-2+3i)\\ &= z^2-2z+3iz+z+2+3i+2iz-4i+6\\ &= z^2-z+5iz-8-i\\ &= z^2+(-1+5i)z+(-8-i)\end{aligned}$$

19 a)

$$(7+2i)(3+8i) = 21+56i+6i-16 = 5+62i$$

19 b)

$$(7-2i)(3-8i) = 21-6i-56i-16 = 5-62i$$

19 c)

$$\begin{aligned} 5^2 + 62^2 &= (5 + 62i)(5 - 62i) \\ &= (7 + 2i)(7 - 2i)(3 + 8i)(3 - 8i) \end{aligned}$$

Considering the real part (the imaginary part is zero):

$$\begin{aligned} &\mathrm{Re}((7 + 2i)(7 - 2i)(3 + 8i)(3 - 8i)) \\ &= (49 + 4)(9 + 64) = 53 \times 73 \end{aligned}$$

20 a)

Let $z = a + bi$ so $z^2 = a^2 - b^2 + 2abi$

$$a^2 - b^2 + 2abi = 5 + 12i$$

Compare real parts: $a^2 - b^2 = 5$

Compare imaginary parts: $2ab = 12$

Solve simultaneously:

$$\begin{aligned} b = \frac{6}{a} \text{ gives } a^2 - \left(\frac{6}{a}\right)^2 &= 5 \\ \Rightarrow a^4 - 36 = 5a^2 &\Rightarrow a^4 - 5a^2 - 36 = 0 \\ a^2 &= \frac{5 \pm \sqrt{25 + 144}}{2} = \frac{5}{2} \pm \frac{13}{2} \end{aligned}$$

so $a^2 = 9$ or $a^2 = -4$

Since a is real, $a = -3$ or $a = 3$

$\therefore a = -3$, $b = -2$ or $a = 3$, $b = 2$

$z = -3 - 2i$ or $z = 3 + 2i$

20 b)

Let $z = a + bi$ so $z^2 = a^2 - b^2 + 2abi$

$$a^2 - b^2 + 2abi = 21 - 20i$$

Compare real parts: $a^2 - b^2 = 21$

Compare imaginary parts: $2ab = -20$

Solve simultaneously:

$$\begin{aligned} b = -\frac{10}{a} \text{ gives } a^2 - \left(-\frac{10}{a}\right)^2 &= 21 \\ \Rightarrow a^4 - 100 = 21a^2 &\Rightarrow a^4 - 21a^2 - 100 = 0 \\ a^2 &= \frac{21 \pm \sqrt{441 + 400}}{2} = \frac{21}{2} \pm \frac{29}{2} \end{aligned}$$

so $a^2 = 25$ or $a^2 = -4$

Since a is real, $a = -5$ or $a = 5$

$\therefore a = -5$, $b = 2$ or $a = 5$, $b = -2$

$z = -5 + 2i$ or $z = 5 - 2i$

20 c)

Let $z = a + bi$ so $z^2 = a^2 - b^2 + 2abi$

$$a^2 - b^2 + 2abi = \frac{3}{16} - \frac{1}{4}i$$

Compare real parts: $a^2 - b^2 = \frac{3}{16}$

Compare imaginary parts: $2ab = -\frac{1}{4}$

Solve simultaneously:

$$b = -\frac{1}{8a} \text{ gives } a^2 - \left(-\frac{1}{8a}\right)^2 = \frac{3}{16}$$
$$\Rightarrow a^4 - \frac{1}{64} = \frac{3}{16}a^2 \Rightarrow 64a^4 - 12a^2 - 1 = 0$$
$$a^2 = \frac{12 \pm \sqrt{144 + 256}}{128} = \frac{12}{128} \pm \frac{20}{128}$$

so $a^2 = \frac{1}{4}$ or $a^2 = -\frac{1}{16}$

Since a is real, $a = -\frac{1}{2}$ or $a = \frac{1}{2}$

$\therefore a = -\frac{1}{2}$, $b = \frac{1}{4}$ or $a = \frac{1}{2}$, $b = -\frac{1}{4}$

$z = -\frac{1}{2} + \frac{1}{4}i$ or $z = \frac{1}{2} - \frac{1}{4}i$

21 a)

$$z = \sqrt{4 + 3i} \Rightarrow z^2 = 4 + 3i$$

Let $z = a + bi$ so $z^2 = a^2 - b^2 + 2abi$

$$a^2 - b^2 + 2abi = 4 + 3i$$

Compare real parts: $a^2 - b^2 = 4$

Compare imaginary parts: $2ab = 3$

Solve simultaneously:

$$b = \frac{3}{2a} \text{ gives } a^2 - \left(\frac{3}{2a}\right)^2 = 4$$
$$\Rightarrow a^4 - \frac{9}{4} = 4a^2 \Rightarrow 4a^4 - 16a^2 - 9 = 0$$
$$a^2 = \frac{16 \pm \sqrt{256 + 144}}{8} = 2 \pm \frac{20}{8}$$

so $a^2 = \frac{9}{2}$ or $a^2 = -\frac{1}{2}$

Since a is real, $a = \frac{3\sqrt{2}}{2}$ or $a = -\frac{3\sqrt{2}}{2}$

$\therefore a = \frac{3\sqrt{2}}{2}$, $b = \frac{\sqrt{2}}{2}$ or $a = -\frac{3\sqrt{2}}{2}$, $b = -\frac{\sqrt{2}}{2}$

$$\therefore \sqrt{4 + 3i} = \frac{3\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

21 b)

Let $z = a + bi$ so $z^2 = a^2 - b^2 + 2abi$

$$a^2 - b^2 + 2abi = -12 + 16i$$

Compare real parts: $a^2 - b^2 = -12$

Compare imaginary parts: $2ab = 16$

Solve simultaneously:

$$b = \frac{8}{a} \text{ gives } a^2 - \left(\frac{8}{a}\right)^2 = -12$$
$$\Rightarrow a^4 - 64 = -12a^2 \Rightarrow a^4 + 12a^2 - 64 = 0$$
$$a^2 = \frac{-12 \pm \sqrt{144 + 256}}{2} = -6 \pm 10$$

so $a^2 = 4$ or $a^2 = -16$

Since a is real, $a = 2$ or $a = -2$

$\therefore a = 2$, $b = 4$ or $a = -2$, $b = -4$

$$\therefore \sqrt{-12 + 16i} = 2 + 4i$$

21 c)

$$z = \sqrt{2 + 4\sqrt{2}i} \Rightarrow z^2 = 2 + 4\sqrt{2}i$$

Let $z = a + bi$ so $z^2 = a^2 - b^2 + 2abi$

$$a^2 - b^2 + 2abi = 2 + 4\sqrt{2}i$$

Compare real parts: $a^2 - b^2 = 2$

Compare imaginary parts: $2ab = 4\sqrt{2}$

Solve simultaneously:

$$b = \frac{2\sqrt{2}}{a} \text{ gives } a^2 - \left(\frac{2\sqrt{2}}{a}\right)^2 = 2$$
$$\Rightarrow a^4 - 8 = 2a^2 \Rightarrow a^4 - 2a^2 - 8 = 0$$
$$a^2 = \frac{2 \pm \sqrt{4 + 32}}{2} = 1 \pm 3$$

so $a^2 = 4$ or $a^2 = -2$

Since a is real, $a = 2$ or $a = -2$

$\therefore a = 2$, $b = \sqrt{2}$ or $a = -2$, $b = -\sqrt{2}$

$$\therefore \sqrt{2 - i} = 2 + \sqrt{2}i$$

22)

Let $z = a + bi$ and $w = c + di$

$$2i(a + bi) + 4(c + di) = -6 + 8i$$

$\Rightarrow 2a + 4d = 8$ and $-2b + 4c = -6$

$$-3(a + bi) + 5i(c + di) = 7 - i$$

$\Rightarrow -3a - 5d = 7$ and $-3b + 5c = -1$

So

$2a + 4d = 8$ and $-3a - 5d = 7$ gives $a = -34$ and $d = 19$

$-2b + 4c = -6$ and $-3b + 5c = -1$ gives $b = -13$ and $c = -8$

So $z = -34 - 13i$ and $w = -8 + 19i$

Alternative Method:

Let $z = a + bi$ and $w = c + di$

$$2i(a + bi) + 4(c + di) = -6 + 8i \quad (1)$$
$$-3(a + bi) + 5i(c + di) = 7 - i \quad (2)$$

Multiply (1) by 3

$$6i(a + bi) + 12(c + di) = -18 + 24i \quad (3)$$

Multiply (2) by $2i$

$$-6i(a + bi) - 10(c + di) = 2 + 14i \quad (4)$$

(3) + (4)

$$2(c + di) = -16 + 38i$$

$\therefore c = -8$ and $d = 19$

Multiply (1) by 5

$$10i(a + bi) + 20(c + di) = -30 + 40i \quad (5)$$

Multiply (2) by $4i$

$$-12i(a + bi) - 20(c + di) = 4 + 28i \quad (6)$$

(5) + (6)

$$-2i(a + bi) = -26 + 68i$$
$$-2ai + 2b = -26 + 68i$$

$\therefore a = -34$ and $b = -13$

So $z = -34 - 13i$ and $w = -8 + 19i$

23 a)

$$(1 + ai)^3 = 1 + 3ai + 3(ai)^2 + (ai)^3 = 1 + 3ai - 3a^2 - a^3i$$
$$\therefore \mathrm{Re}((1 + ai)^3) = 1 - 3a^2$$
$$1 - 3a^2 = 0 \Rightarrow 3a^2 = 1 \Rightarrow a^2 = \frac{1}{3} \Rightarrow a = \pm\sqrt{\frac{1}{3}} = \pm\frac{\sqrt{3}}{3}$$

23 b)

$$(3 + ai)^3 = 27 + 27ai + 9(ai)^2 + (ai)^3 = 27 + 27ai - 9a^2 - a^3i$$
$$\therefore \mathrm{Im}((3 + ai)^3) = 27a - a^3$$
$$27a - a^3 = 0 \Rightarrow a(27 - a^2) = 0 \Rightarrow a = 0 \text{ or } a = \pm 3\sqrt{3}$$

23 c)

$$\begin{aligned}(4+ai)^4 &= 16+32ai+24(ai)^2+8(ai)^3+(ai)^4\\ &= 256+256ai-96a^2-16a^3i+a^4\\ \therefore \mathrm{Im}((4+ai)^4) &= 256a-16a^3\end{aligned}$$

$$256a-16a^3=0 \Rightarrow 16a(16-a^2)=0 \ \therefore a=0,\ a=-4 \text{ or } a=4$$

23 d)

$$\begin{aligned}(2-ai)^4 &= 16-32ai+24(-ai)^2+8(-ai)^3+(-ai)^4\\ &= 16-32ai-24a^2+8a^3i+a^4\\ \therefore \mathrm{Re}((2-ai)^4) &= 16-24a^2+a^4\end{aligned}$$

$$a^4-24a^2+16=0 \Rightarrow a^2=\frac{24\pm\sqrt{576-64}}{2}=12\pm 8\sqrt{2}$$

$$\therefore a=\sqrt{12+8\sqrt{2}},\ a=-\sqrt{12+8\sqrt{2}},\ a=\sqrt{12-8\sqrt{2}} \text{ or } a=-\sqrt{12-8\sqrt{2}}$$

23 e)

$$\begin{aligned}(1-ai)^5 &= 1+5ai+10(-ai)^2+10(-ai)^3+5(-ai)^4+(-ai)^5\\ &= 1+5ai-10a^2+10a^3i+5a^4-a^5i\\ \therefore \mathrm{Im}((1-ai)^5) &= 5a+10a^3-a^5\end{aligned}$$

$$5a+10a^3-a^5=0 \Rightarrow -a(a^4-10a^2-5)=0$$

$$\Rightarrow \text{so } a=0 \text{ or } a^2=\frac{10\pm\sqrt{100+20}}{2}=5\pm\sqrt{30}$$

$$\therefore a=0,\ a=\sqrt{5+\sqrt{30}} \text{ or } a=\sqrt{5-\sqrt{30}}$$

24 a)

$$z^*=2-i$$

24 b)

$$z^*=3+i$$

24 c)

$$z^*=-6+i$$

24 d)

$$z^*=1-5i$$

24 e)

$$z^* = -4 + 2i$$

24 f)

$$z^* = \frac{1}{2} - \frac{3}{4}i$$

24 g)

$$z^* = \sqrt{5} + \sqrt{3}i$$

24 h)

$$z^* = -2i$$

24 i)

$$z^* = -10$$

25 a)

$$z + z^* = a + bi + (a - bi) = 2a \quad \text{which is real.}$$

25 b)

$$zz^* = (a + bi)(a - bi) = a^2 + abi - abi + b^2 = a^2 + b^2 \quad \text{which is real.}$$

26 a)

$$2w - z = 2(2 - i) - (-5 + 3i) = 4 - 2i + 5 - 3i = 9 - 5i$$

26 b)

$$w^* + w = (2 + i) + (2 - i) = 4$$

26 c)

$$2w^* + z = 2(2 + i) + (-5 + 3i) = 4 + 2i - 5 + 3i = -1 + 5i$$

26 d)

$$z^2 = (-5+3i)^2 = (-5+3i)(-5+3i) = 25 - 15i - 15i - 9 = 16 - 30i$$

26 e)

$$\begin{aligned} wzz^* &= (2-i)(-5+3i)(-5-3i) = (2-i)(25-15i+15i+9) \\ &= (2-i) \times 34 \\ &= 68 - 34i \end{aligned}$$

26 f)

$$\begin{aligned} w(z^*)^2 &= (2-i)(-5-3i)^2 = (2-i)(25+15i+15i-9) \\ &= (2-i)(16+30i) = 32 + 60i - 16i + 30 \\ &= 62 + 44i \end{aligned}$$

26 g)

$$\begin{aligned} (w^*)^3 &= (2+i)^3 = (2+i)(2+i)(2+i) = (2+i)(4+4i-1) \\ &= (2+i)(3+4i) = 6 + 8i + 3i - 4 = 2 + 11i \end{aligned}$$

27 a)

$$\begin{aligned} 4iz + 3z^* &= 4i(x+iy) + 3(x-iy) = 4xi - 4y + 3x - 3iy \\ &= (3x-4y) + (4x-3y)i \\ \mathrm{Re}(4iz+3z^*) &= 3x - 4y \\ \mathrm{Im}(4iz+3z^*) &= 4x - 3y \end{aligned}$$

27 b)

$3x - 4y = 5$ and $4x - 3y = -3$

Solving simultaneously gives $x = -\frac{27}{7}$ and $y = -\frac{29}{7}$

28 a)

$$2z^* = 3 - 4i \Rightarrow z^* = \frac{3}{2} - 2i \ \ \therefore z = \frac{3}{2} + 2i$$

28 b)

$$3z^* = -1 + 8i \Rightarrow z^* = -\frac{1}{3} + \frac{8}{3}i \ \ \therefore z = -\frac{1}{3} - \frac{8}{3}i$$

28 c)

$$iz^* = -1 + i \Rightarrow -i \times iz^* = -i(-1+i) = 1 + i \ \ \therefore z = 1 - i$$

Alternative Method:

$$z^* = \frac{-1+i}{i} \times \frac{-i}{-i} = -i(-1+i) = 1 + i \ \ \therefore z = 1 - i$$

29 a)

Let $z = a + bi$

$$z + 2z^* = 3 + 9i \Rightarrow a + bi + 2(a - bi) = 3 + 9i$$

Compare real parts: $a + 2a = 3 \ \ \therefore a = 1$

Compare imaginary parts: $b - 2b = 9 \ \ \therefore b = -9$

$\therefore z = 1 - 9i$

29 b)

Let $z = a + bi$

$$2z - z^* = -4 - 5i \Rightarrow 2(a + bi) - (a - bi) = -4 - 5i$$

Compare real parts: $2a - a = -4 \ \ \therefore a = -4$

Compare imaginary parts: $2b + b = -5 \ \ \therefore b = -\frac{5}{3}$

$\therefore z = -4 - \frac{5}{3}i$

29 c)

Let $z = a + bi$

$$z + 5z^* = -i \Rightarrow a + bi + 5(a - bi) = -i$$

Compare real parts: $a + 5a = 0 \ \ \therefore a = 0$

Compare imaginary parts: $b - 5b = -1 \ \ \therefore b = \frac{1}{4}$

$\therefore z = \frac{1}{4}i$

30 a)

Let $z = a + bi$ so $z^* = a - bi$

$$(z^*)^2 = (a - bi)^2 = a^2 - b^2 - 2abi$$
$$(z^2)^* = ((a + bi)(a + bi))^* = (a^2 - b^2 + 2abi)^* = a^2 - b^2 - 2abi$$

$\therefore (z^*)^2 = (z^2)^*$ as req.

30 b)

Let $z = a + bi$ so $z^* = a - bi$

$$\begin{aligned}(z^*)^3 &= (a - bi)^2(a - bi) = (a^2 - b^2 - 2abi)(a - bi) \\ &= a^3 - ab^2 - 2a^2bi - a^2bi + b^3i - 2ab^2 \\ &= a^3 - 3ab^2 + (b^3 - 3a^2b)i \\ (z^3)^* &= ((a + bi)^2(a + bi))^* = ((a^2 - b^2 + 2abi)(a + bi))^* \\ &= (a^3 - ab^2 + 2a^2bi + a^2bi - b^3i - 2ab^2)^* \\ &= (a^3 - ab^2 - 2ab^2 + (-b^3 + a^2b)i)^* \\ &= a^3 - ab^2 - 2ab^2 + (b^3 - a^2b)i\end{aligned}$$

$\therefore (z^*)^3 = (z^3)^*$ as req.

31 a)

$$z - z^* = a + bi - (a - bi) = 2bi$$

31 b)

$$(z - z^*)^2 = (a + bi - (a - bi))^2 = (2bi)^2 = -4b^2$$

31 c)

$$(z - z^*)^3 = (a + bi - (a - bi))^3 = (2bi)^3 = -8b^3 i$$

31 d)

$$(z - z^*)^4 = (a + bi - (a - bi))^4 = (2bi)^4 = 16b^4$$

31 e)

$$(z - z^*)^{18} = (a + bi - (a - bi))^{18} = (2bi)^{18} = -262144b^{18}$$

32)

Let $z = a + bi$ so $z^* = a - bi$

$$\begin{aligned}(4z - z^*)^* &= (4(a + bi) - (a - bi))^* = (4a + 4bi - a + bi)^* \\ &= (3a + 5bi)^* = 3a - 5bi \\ z^2 &= (a + bi)^2 = a^2 - b^2 + 2abi\end{aligned}$$

Compare real parts: $a^2 - b^2 = 3a$

Compare imaginary parts: $2ab = -5b$

Solve simultaneously:

Consider $2ab = -5b$

If $b \neq 0$, $a = -\frac{5}{2}$ which gives $\left(-\frac{5}{2}\right)^2 - b^2 = 3\left(-\frac{5}{2}\right)$

$$\Rightarrow b^2 = \frac{25}{4} + \frac{15}{2} = \frac{55}{4} \text{ so } b = \frac{\sqrt{55}}{2} \text{ or } b = -\frac{\sqrt{55}}{2}$$

$\therefore a = -\frac{5}{2}$, $b = \frac{\sqrt{55}}{2}$ or $a = -\frac{5}{2}$, $b = -\frac{\sqrt{55}}{2}$

If $b = 0$, $a^2 = 3a \Rightarrow a^2 - 3a = 0 \Rightarrow a(a - 3) = 0$

So $a = 0$ or $a = 3$

$\therefore a = 0$, $b = 0$ or $a = 3$, $b = 0$

Which gives

$$z = 0,\ z = 3,\ z = -\frac{5}{2} + \frac{\sqrt{55}}{2}i \text{ or } z = -\frac{5}{2} - \frac{\sqrt{55}}{2}i$$

33)

Let $z = a + bi$, $z^* = a - bi$ and $w = c + di$

$$2(a + bi) + 4(a - bi) = 18 + 4i$$

$\Rightarrow 2a + 4a = 18$ so $a = 3$ and $2b - 4b = 4$ so $b = -2$

$z = 3 - 2i$

So $z^* = 3 + 2i$

$$5(c + di) - 3(3 + 2i) = -3 + 2i$$

$\Rightarrow 5c - 9 = -3$, so $c = \frac{6}{5}$ and $5d - 6 = 2$ so $d = \frac{8}{5}$

$$\therefore w = \frac{6}{5} + \frac{8}{5}i$$

34)

Let $z = a + bi$

$$(a + bi)^2 + 3i(a - bi) = 8$$
$$a^2 - b^2 + 2abi + 3ai + 3b = 8$$

Compare real part: $a^2 - b^2 + 3b = 8$

Compare imaginary part: $2ab + 3a = 0 \Rightarrow a(2b + 3) = 0$

Since $\mathrm{Re}(z) = a < 0$, $b = -\frac{3}{2}$

$$a^2 - \left(-\frac{3}{2}\right)^2 + 3\left(-\frac{3}{2}\right) = 8 \Rightarrow a^2 = \frac{59}{4}$$

$\therefore a = \frac{\sqrt{59}}{2}$ or $a = -\frac{\sqrt{59}}{2}$

$$\therefore z = -\frac{\sqrt{59}}{2} - \frac{3}{2}i$$

35)

Let $z = a + bi$

$$(a + bi)^2 + 4i(a - bi) = 10i$$
$$a^2 - b^2 + 2abi + 4a - 4bi = 10i$$

Compare real part: $a^2 - b^2 + 4a = -4$

Compare imaginary part: $2ab - 4b = 10 \Rightarrow 2b(a - 2) = 10 \Rightarrow b = \frac{5}{a-2}$

So that $a^2 - b^2 + 4a = -4$ becomes:

$$a^2 - \left(\frac{5}{a-2}\right)^2 + 4a = -4$$
$$a^2(a-2)^2 - 25 + 4a(a-2)^2 + 4(a-2)^2 = 0$$
$$a^2(a^2 - 4a + 4) - 25 + 4a(a^2 - 4a + 4) + 4(a^2 - 4a + 4) = 0$$
$$a^4 - 4a^3 + 4a^2 - 25 + 4a^3 - 16a^2 + 16a + 4a^2 - 16a + 16 = 0$$
$$a^4 - 8a^2 - 9 = 0$$
$$(a^2 + 1)(a^2 - 9) = 0$$

Since a is real $a = \pm 3$

But also $\mathrm{Re}(z) = a > 0$ so $a = 3$

$$b = \frac{5}{3-2} = 5$$

$\therefore z = 3 + 5i$

36 a)

$$\frac{4}{i} \times \frac{-i}{-i} = -4i$$

36 b)

$$\frac{7}{1+i} \times \frac{1-i}{1-i} = \frac{7-7i}{1+1} = \frac{7}{2} - \frac{7}{2}i$$

36 c)

$$\frac{9}{5-2i} \times \frac{5+2i}{5+2i} = \frac{45+18i}{25+4} = \frac{45}{29} + \frac{18}{29}i$$

36 d)

$$\frac{8}{2+3i} \times \frac{2-3i}{2-3i} = \frac{16-24i}{4+9} = \frac{16}{13} - \frac{24}{13}i$$

36 e)

$$\frac{2+i}{4-i} \times \frac{4+i}{4+i} = \frac{8+6i-1}{16+1} = \frac{7}{17} + \frac{6}{17}i$$

36 f)

$$\frac{(1-i)^2}{1+i} = \frac{-2i}{1+i} \times \frac{1-i}{1-i} = \frac{-2i-2}{1+1} = -1-i$$

36 g)

$$\begin{aligned}\frac{a+bi}{b-ai} \times \frac{b+ai}{b+ai} &= \frac{ab+a^2i+b^2i-ab}{b^2+a^2} \\ &= \frac{a^2i+b^2i}{b^2+a^2} = \frac{(a^2+b^2)i}{b^2+a^2} = i\end{aligned}$$

37)

$$\begin{aligned}\frac{5}{z} &= \frac{5}{-1+2i} \times \frac{-1-2i}{-1-2i} = \frac{-5-10i}{1+4} = -1-2i \\ z + \frac{5}{z} &= -1+2i+\frac{5}{-1+2i} = -1+2i-1-2i = -2\end{aligned}$$

38 a)

$$(3-2i)(4+i) = 12 - 5i + 2 = (14 - 5i) \text{ volts}$$

38 b)

$$\frac{6-2i}{2+3i} \times \frac{2-3i}{2-3i} = \frac{12 - 18i - 4i - 6}{4+9} = \left(\frac{6}{13} - \frac{22}{13}i\right) \text{ ohms}$$

39)

$$\begin{aligned}\frac{15-5i}{z} = 2 - 3i \Rightarrow z &= \frac{15-5i}{2-3i} \times \frac{2+3i}{2+3i} \\ &= \frac{30 - 10i + 45i + 15}{4+9} = \frac{45}{13} + \frac{35}{13}i\end{aligned}$$

40)

$$\frac{2}{3+i} \times \frac{3-i}{3-i} = \frac{6-2i}{9+1} = \frac{3}{5} - \frac{1}{5}i$$

and

$$\frac{3}{2+i} \times \frac{2-i}{2-i} = \frac{6-3i}{4+1} = \frac{6}{5} - \frac{3}{5}i$$

So

$$\frac{2}{3+i} + \frac{3}{2+i} = \frac{3}{5} - \frac{1}{5}i + \frac{6}{5} - \frac{3}{5}i = \frac{9}{5} - \frac{4}{5}i$$

41)

$$\begin{aligned}\frac{1}{\cos\theta - i\sin\theta} \times \frac{\cos\theta + i\sin\theta}{\cos\theta + i\sin\theta} &= \frac{\cos\theta + i\sin\theta}{\cos^2\theta + \sin^2\theta} \\ &= \cos\theta + i\sin\theta\end{aligned}$$

42 a)

$$\frac{1}{x+iy} \times \frac{x-iy}{x-iy} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i$$

$\therefore \text{Re}\left(x + iy + \frac{1}{x+iy}\right) = x + \frac{x}{x^2+y^2}$ and $\text{Im}\left(x + iy + \frac{1}{x+iy}\right) = y - \frac{y}{x^2+y^2}$

42 b)

$$-\frac{1}{x+iy} \times \frac{x-iy}{x-iy} = -\frac{x-iy}{x^2+y^2} = -\frac{x}{x^2+y^2} + \frac{y}{x^2+y^2}i$$

$\therefore \text{Re}\left(x + iy - \frac{1}{x+iy}\right) = x - \frac{x}{x^2+y^2}$ and $\text{Im}\left(x + iy - \frac{1}{x+iy}\right) = y + \frac{y}{x^2+y^2}$

42 c)

$$\frac{1}{x-iy}\times\frac{x+iy}{x+iy}=\frac{x+iy}{x^2+y^2}=\frac{x}{x^2+y^2}+\frac{y}{x^2+y^2}i$$

$\therefore \text{Re}\left(x-iy+\frac{1}{x-iy}\right)=x+\frac{x}{x^2+y^2}$ and $\text{Im}\left(x-iy+\frac{1}{x-iy}\right)=-y+\frac{y}{x^2+y^2}$

42 d)

$$-\frac{1}{x-iy}\times\frac{x+iy}{x+iy}=-\frac{x+iy}{x^2+y^2}=-\frac{x}{x^2+y^2}-\frac{y}{x^2+y^2}i$$

$\therefore \text{Re}\left(x-iy-\frac{1}{x-iy}\right)=x-\frac{x}{x^2+y^2}$ and $\text{Im}\left(x-iy-\frac{1}{x-iy}\right)=-y-\frac{y}{x^2+y^2}$

43)

$$\frac{1+i}{1-i}=\frac{1+i}{1-i}\times\frac{1+i}{1+i}=\frac{1+2i-1}{1+1}=i$$

So least positive integer n such that $i^n=1$

$$i^2=-1,\ i=-i,\ i^4=1\ \ \therefore n=4$$

44)

$$\frac{a(1-2i)(3-2i)}{1-2i}+\frac{b(1-2i)(3-2i)}{3-2i}=\frac{-5+6i}{1+8i}\times(1-2i)(3-2i)$$
$$a(3-2i)+b(1-2i)=\frac{-5+6i}{1+8i}\times(-1-8i)$$
$$(3a+b)+(-2a-2b)i=5-6i$$

$3a+b=5$ and $-2a-2b=-6$

Which gives $a=1$, $b=2$

Alternative Method:

$$\frac{a}{1-2i}\times\frac{1+2i}{1+2i}=\frac{a+2ai}{1+4}=\frac{a}{5}+\frac{2a}{5}i$$
$$\frac{b}{3-2i}\times\frac{3+2i}{3+2i}=\frac{3b+2bi}{9+4}=\frac{3b}{13}+\frac{2b}{13}i$$
$$\frac{-5+6i}{1+8i}\times\frac{1-8i}{1-8i}=\frac{-5+40i+6i+48}{1+64}=\frac{43}{65}+\frac{46}{65}i$$

$\frac{a}{5}+\frac{3b}{13}=\frac{43}{65}$ and $\frac{2a}{5}+\frac{2b}{13}=\frac{46}{65}$

Solving simultaneously: $a=1$, $b=2$

45)

$$zz^*=(a+bi)(a-bi)=a^2+b^2=17$$
$$\frac{z}{z^*}=\frac{a+bi}{a-bi}\times\frac{a+bi}{a+bi}=\frac{a^2-b^2+2abi}{a^2+b^2}$$
$$=\frac{a^2-b^2}{a^2+b^2}+\frac{2ab}{a^2+b^2}i$$

so that $\frac{a^2-b^2}{a^2+b^2} = \frac{15}{17}$ and $\frac{2ab}{a^2+b^2} = -\frac{8}{17}$

So $a^2 - b^2 = 15$ and $2ab = -8$

Solve simultaneously:

$$b = -\frac{4}{a} \text{ gives } a^2 - \left(-\frac{4}{a}\right)^2 = 15$$
$$\Rightarrow a^4 - 16 = 15a^2 \Rightarrow a^4 - 15a^2 - 16 = 0$$
$$a^2 = \frac{15 \pm \sqrt{225+64}}{2} = \frac{15}{2} \pm \frac{17}{2}$$

so $a^2 = 16$ or $a^2 = -1$

Since a is real, $a = 4$ or $a = -4$

$\therefore a = 4$, $b = -1$ or $a = -4$, $b = 1$

46)

Let $z = a + bi$

$$\frac{z}{z-2} = \frac{a+bi}{a-2+bi} \times \frac{a-2-bi}{a-2-bi}$$
$$= \frac{a^2 - 2a - abi + abi - 2bi + b^2}{(a-2)^2 + b^2}$$
$$= \frac{a^2 - 2a - 2bi + b^2}{(a-2)^2 + b^2}$$
$$= \frac{a^2 - 2a + b^2}{(a-2)^2 + b^2} + \frac{-2b}{(a-2)^2 + b^2}i$$
$$\text{Re}\left(\frac{z}{z-2}\right) = \frac{a^2 - 2a + b^2}{(a-2)^2 + b^2}$$
$$\text{Im}\left(\frac{z}{z-2}\right) = \frac{-2b}{(a-2)^2 + b^2}$$

47)

$$\frac{z}{3+4i} = \frac{a+bi}{3+4i} \times \frac{3-4i}{3-4i} = \frac{3a - 4ai + 3bi + 4b}{9+16}$$
$$= \frac{3a+4b}{25} + \frac{-4a+3b}{25}i$$
$$\frac{z-1}{5i} = \frac{a+bi-1}{5i} \times \frac{-5i}{-5i} = \frac{-5ai + 5b + 5i}{25}$$
$$= \frac{5b}{25} + \frac{-5a+5}{25}i$$
$$\frac{5}{3-4i} \times \frac{3+4i}{3+4i} = \frac{15+20i}{9+16} = \frac{15}{25} + \frac{20}{25}i$$

Equating real and imaginary parts:

$$\frac{3a+4b}{25} + \frac{5b}{25} = \frac{15}{25} \quad \text{and} \quad \frac{-4a+3b}{25} + \frac{-5a+5}{25} = \frac{20}{25}$$

$3a + 9b = 15$ and $-9a + 3b + 5 = 20$

$\therefore a = -1$, $b = 2$

$\therefore z = -1 + 2i$

48)

$$\begin{aligned}\frac{a-bi}{c-di} &= \frac{a-bi}{c-di}\times\frac{c+di}{c+di} = \frac{ac-bci+adi+bd}{c^2+d^2}\\ &= \frac{ac+bd}{c^2+d^2} - \frac{bc-ad}{c^2+d^2}i\\ (x-iy)^2 &= \frac{a-bi}{c+di} = \frac{ac+bd}{c^2+d^2} - \frac{bc-ad}{c^2+d^2}i\\ (x-iy)^2 &= (x-iy)(x-iy) = x^2-y^2-2xyi\end{aligned}$$

Compare real components:

$$x^2 - y^2 = \frac{ac+bd}{c^2+d^2} \quad \text{as req.}$$